

KITEWRIGHT LIFT · DESIGN DECISION RECORD

Kitewright Lift R4 restated to the rated payload (open decision 4 closed)

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DOCUMENT CONTROL

Document ID	KWL-DDR-005	Version	0.1
Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-lift		
Source	docs/decisions/0005-r4-rated-payload.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-04	Amish Chadha	Open decision 4 closed as a consequence of Amish's round-3 decision 8A

0005: R4 restated to the rated payload

- **Date:** 2026-10-04
- **Status:** accepted. Decided by Amish Chadha on 2026-10-04: "For round 3, I agree with all your proposed recommendations". Open decision 4 follows from round-3 decision 8A (the payload rated at what fits inside the 25 kg take-off limit) and is closed with its recommended option A.

■ Context

R4 asked for at least 20 min of hover "with 5 kg at sea level". Under decision 8A (KWL-DDR-004) Lift keeps its 25 kg take-off limit (R1) with two ColdCell packs of 4.46 kg and rates its payload at what fits, about 3.3 kg (R2, 3.2 kg on paper). A 5 kg payload no longer fits inside R1, so the sea-level case was being read at the rated payload.

■ Decision

R4 restated in KWL-REQ-001 v0.5: "At least 20 min with the rated payload (R2) at sea level and 10 min with 2 kg at 5,000 m and -20 °C (targets)". The altitude case is unchanged.

■ Results (KWL-CAL-001 v0.3, unchanged)

20.4 min at sea level with the rated 3.2 kg (25.0 kg take-off) and 14.5 min at 5,000 m and -20 °C with 2 kg. **R4 met on paper.**

■ Consequences

- Records only: no change to the model, calculations, BOM, drawings or pictures.
- A heavier sea-level payload would need a lighter airframe or packs in a later prototype; that is not a requirement of this one.

Safety: The rated payload is the most Lift may carry; flying above it breaks R1's 25 kg limit and the class rules that rest on it.